Das Kiwitsche 2-Punkt-Verfahren: Bang-Bang Control as a Starting Point for Optimal PID Parameter Estimation

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Abstract

The *Kiwitsche 2-Punkt-Verfahren* (KZV) is a practical, four-phase procedure that uses an existing Bang-Bang (two-point) controller to identify an FOPDT plant model and synthesise an IMC-based PID controller — all without requiring a separate open-loop experiment. The method runs the Bang-Bang controller until a stationary limit cycle is established, extracts the oscillation period and amplitude, and uses transient-slope measurements to estimate the process gain K, time constant T, and dead time L. The identified model is then fed into the Rivera–Morari IMC-PID tuning rules to obtain the gains $(K_\mathrm{p}, T_\mathrm{i}, T_\mathrm{d})$. Simulations on a benchmark FOPDT plant show that the resulting controller achieves 5.5 % overshoot and an IAE within 2 % of the ideal IMC result, versus 57.9 % overshoot and an eightfold larger IAE for classical Ziegler–Nichols tuning.

Keywords: PID tuning, Bang-Bang control, two-point control, parameter estimation, FOPDT identification, limit cycle, industrial cooling, IMC.

1 Introduction

PID controllers account for more than 90% of deployed control loops in industrial process control [2], yet selecting appropriate gains remains a practical challenge. Classical tuning rules — Ziegler–Nichols (ZN) [9], Cohen–Coon [5], IMC [6] — require an open-loop step test or explicit prior knowledge of the process. In industrial settings such tests are often infeasible due to production constraints or safety requirements.

Bang-Bang (two-point) controllers are already widely deployed as thermostats and safety limiters in cooling, heating, and refrigeration plants. They inherently drive the closed loop into a limit-cycle oscillation that encodes the plant dynamics. Figure 1 illustrates this: the duty cycle shifts with setpoint, reflecting the steady-state gain, while the waveform shape carries information about the time constant and dead time.

1.1 The KZV Approach

This paper presents the *Kiwitsche 2-Punkt-Verfahren* (KZV): a four-phase procedure that exploits the Bang-Bang limit cycle to identify the plant's FOPDT model and synthesise an IMC-PID controller without any additional open-loop experiment. The KZV is a specialised variant of the relay-feedback idea of Åström and Hägglund [1]: instead of reading a single Nyquist point, it uses time-domain transient slopes within each half-cycle to extract the full FOPDT triple (K, T, L). Figure 2 shows the phase-plane orbits produced by the same plant at three setpoints — each orbit is a reproducible fingerprint of the local plant dynamics.

Section 2 reviews the theoretical foundations; section 3 details the KZV procedure; section 4 presents simulation results; and section 6 concludes.

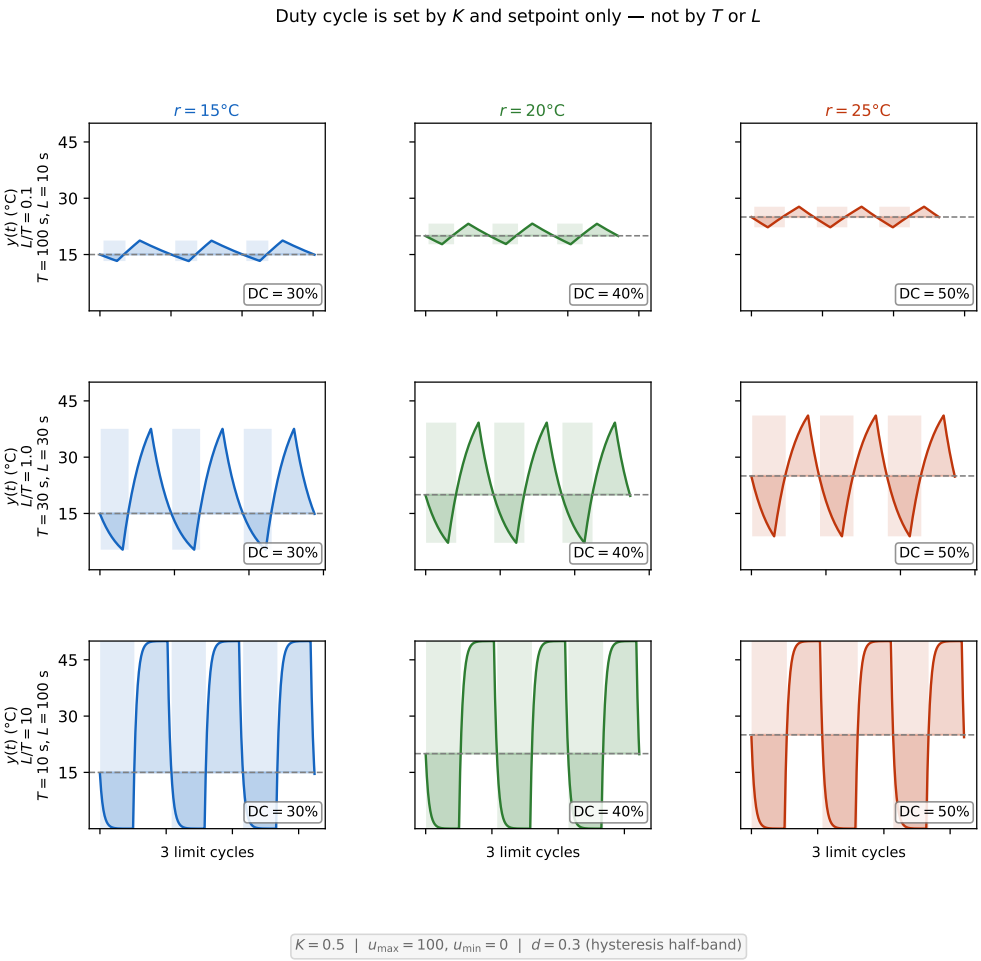

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Figure 1: Bang-Bang limit cycles at three setpoints ($r \in \{15, 20, 25\}$ °C) for three FOPDT plants sharing the same gain K but different L/T ratios: lag-dominated ($L/T = 0.1$, $T = 100\,\mathrm{s}$, $L = 10\,\mathrm{s}$), balanced ($L/T = 1.0$, $T = 30\,\mathrm{s}$, $L = 30\,\mathrm{s}$), and dead-time dominant ($L/T = 10$, $T = 10\,\mathrm{s}$, $L = 100\,\mathrm{s}$). Each panel shows the plant output $y(t)$ with the ON-phase tinted and the duty cycle (DC) annotated. The DC values are identical across all three plants at every setpoint, confirming that duty cycle depends only on K and r — not on T or L. The oscillation period and waveform shape do differ, which is why the KZV extracts T and L separately.

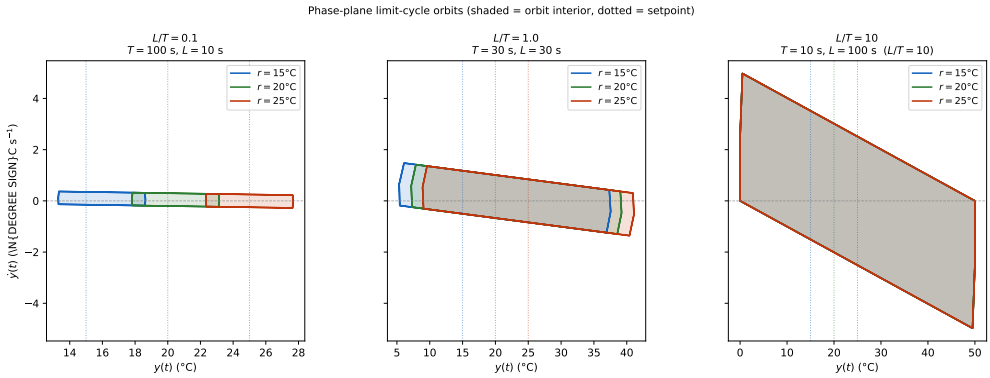

Figure 2: Phase-plane limit-cycle orbits (y vs. $\dot{y}$) at $r \in \{15, 20, 25\}$ °C. Each closed curve encodes the local FOPDT parameters at that operating point.

2 Theoretical Background

2.1 Bang-Bang Control and Limit Cycles

A Bang-Bang (two-point) controller with hysteresis d generates the manipulated variable signal

$$u(t) = \begin{cases} U_{\max} & \text{if } e(t) > +d, \\ U_{\min} & \text{if } e(t) < -d, \\ u(t^-) & \text{otherwise,} \end{cases} \tag{1}$$

where $e(t) = r - y(t)$ is the control error, r is the setpoint, $y(t)$ is the plant output, and $u(t^-)$ denotes the previous value (memory).

Figure 3 illustrates the closed-loop structure. For a stable plant $G(s)$ with sufficient gain and phase delay, this loop converges to a periodic *limit-cycle oscillation* [3]. Let T_{u} be the oscillation period and A_{u} the peak-to-peak amplitude of $y(t)$ in steady limit-cycle operation.

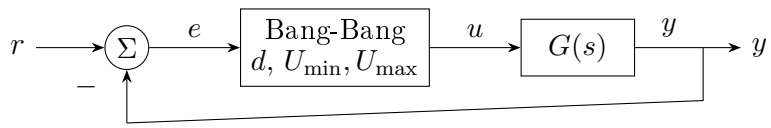

Figure 3: Closed-loop block diagram: Bang-Bang controller drives the plant $G(s)$ into a sustained limit-cycle oscillation around the setpoint r.

2.2 Describing Function Analysis

The describing function (DF) of a relay with hysteresis d and amplitude M (where $M = (U_{\max} - U_{\min})/2$) is [4]

$$N(A) = \frac{4M}{\pi A}\left(\sqrt{1-\left(\frac{d}{A}\right)^2} - j\frac{d}{A}\right), \quad A > d, \tag{2}$$

where $A = A_{\mathrm{u}}/2$ is the single-sided oscillation amplitude of the output.

The limit-cycle frequency $\omega_u = 2\pi/T_{\mathrm{u}}$ satisfies the harmonic balance condition

$$G(j\omega_u) = -\frac{1}{N(A)}. \tag{3}$$

For the special case of zero hysteresis ($d = 0$), (2) reduces to the ideal relay describing function $N(A) = 4M/(\pi A)$, and (3) becomes equivalent to the Ziegler–Nichols ultimate gain condition $|G(j\omega_u)| = \pi A/(4M)$.

2.3 FOPDT Plant Model

The First-Order-Plus-Dead-Time (FOPDT) transfer function

$$G(s) = \frac{Ke^{-Ls}}{Ts+1} \tag{4}$$

is a standard approximation for a large class of industrial processes. It is characterised by three parameters: process gain K, time constant T, and dead time L. Figure 4 shows the internal structure of this model.

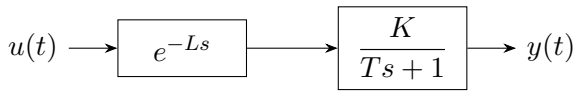

Figure 4: Internal block diagram of the FOPDT model (4): the dead-time element delays the input, which then drives a first-order lag with gain K and time constant T.

2.3.1 Parameter Identification from Limit Cycle

Evaluating (4) at $s = j\omega_u$ gives

$$|G(j\omega_u)| = \frac{K}{\sqrt{1+(\omega_u T)^2}}, \tag{5}$$

$$\angle G(j\omega_u) = -\arctan(\omega_u T) - \omega_u L = -\pi. \tag{6}$$

The describing function provides two equations (gain and phase of $-1/N(A)$):

$$|G(j\omega_u)| = \frac{\pi A}{4M} \frac{1}{\sqrt{1-(d/A)^2}}, \tag{7}$$

$$\angle G(j\omega_u) = -\pi + \arcsin\left(\frac{d}{A}\right). \tag{8}$$

Combining (6) and (8) yields

$$\arctan(\omega_u T) + \omega_u L = \pi - \arcsin\left(\frac{d}{A}\right). \tag{9}$$

With ω_u known from the measured period T_u, (5), (7) and (9) form a system that can be solved for K, T, and L if one additional assumption is introduced. A common engineering choice is to adopt the ratio L/T from a characteristic of the oscillation waveform (e.g., the symmetry of the on/off intervals) or to set $L = T_\mathrm{u}/(2\pi) \cdot \phi_L$ based on the phase margin [7].

Remark 2.1. *For processes where the FOPDT approximation is poor (e.g., resonant systems), higher-order describing function methods or relay feedback with filtered output should be considered.*

Remark 2.2. *While (5), (7) and (9) characterise the limit cycle and guarantee the existence and uniqueness of the FOPDT parameters K, T, L, the KZV does not solve this nonlinear system directly. Instead, Phase 3 (section 3.3) employs a transient-slope method to identify $\hat{L}$, $\hat{T}$, $\hat{K}$ sequentially from the time-domain signal. This approach is more robust in practice because it avoids the numerical sensitivity of the nonlinear DF system while remaining theoretically justified by the DF equations.*

2.4 IMC-Based PID Tuning

Given the FOPDT model (4), the IMC-PID tuning rules [6] yield

$$K_\mathrm{p} = \frac{1}{K} \cdot \frac{T}{\lambda + L}, \tag{10}$$

$$T_\mathrm{i} = T, \tag{11}$$

$$T_\mathrm{d} = \frac{L}{2}, \tag{12}$$

where $\lambda > 0$ is the desired closed-loop time constant and serves as a robustness tuning knob: larger λ gives a more sluggish but more robust response.

A recommended default is $\lambda = L$ (i.e., equal to the estimated dead time). This choice, consistent with the guideline $\lambda \geq 0.5L$ recommended in [6], provides a closed-loop bandwidth approximately equal to the reciprocal of twice the dead time, yielding a good trade-off between speed and robustness for typical industrial processes.

Remark 2.3. *The modified ZN formulas for FOPDT (see, e.g., [2]) are an alternative and are compared against the IMC rules in section 4.*

3 The Kiwitsche 2-Punkt-Verfahren

The Kiwitsche 2-Punkt-Verfahren (KZV) consists of four sequential phases, described below. An overview is shown in fig. 5.

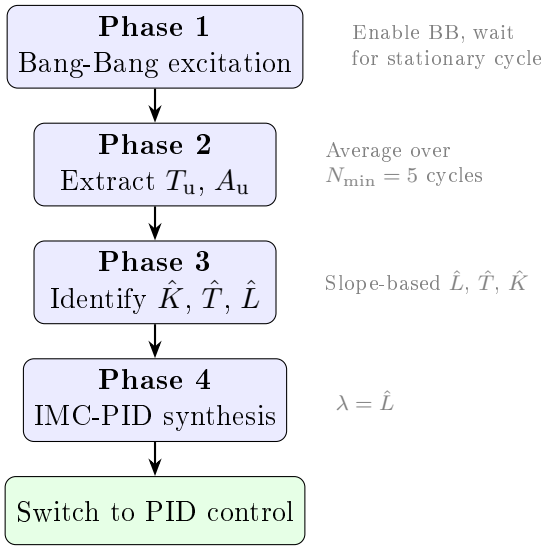

Figure 5: Flow chart of the Kiwitsche 2-Punkt-Verfahren (KZV).

3.1 Phase 1: Bang-Bang Excitation

1. Set the Bang-Bang controller parameters: switching amplitude $M = (U_{\max} - U_{\min})/2$ and hysteresis band d. For most processes, a hysteresis of $d \approx 0.5$–2% of the measurement range is appropriate.

2. Set the setpoint r to a representative operating point (e.g. the desired cooling temperature).

3. Enable the Bang-Bang controller and wait until the output $y(t)$ has settled into a *stationary* limit cycle. Stationarity is verified by checking that three consecutive oscillation periods agree to within $\pm 5\%$.

3.2 Phase 2: Characteristic Extraction

Once a stationary limit cycle is established, let $N \geq N_{\min}$ be the number of recorded cycles, with $N_{\min} = 5$. Record N consecutive oscillation cycles and compute:

$$T_{\mathrm{u}} = \frac{1}{N_{\min}} \sum_{k=1}^{N_{\min}} (t_{k+1} - t_k), \tag{13}$$

$$A_{\mathrm{u}} = \frac{1}{N_{\min}} \sum_{k=1}^{N_{\min}} \big(y_{\max,k} - y_{\min,k}\big), \tag{14}$$

where t_k are the times of successive positive zero-crossings of $e(t)$, and $y_{\max,k}$, $y_{\min,k}$ are the maximum and minimum of $y(t)$ within each cycle.

3.3 Phase 3: Plant Identification

Using the measured quantities $(T_{\mathrm{u}}, A_{\mathrm{u}}, d, M)$ and the recorded time series (t, y, u), the FOPDT parameters are estimated as follows. The three estimates must be computed in the strict order $\hat{L} \to \hat{T} \to \hat{K}$: $\hat{T}$ depends on $\hat{L}$, and $\hat{K}$ depends on $\hat{T}$.

Dead-time estimate. For a FOPDT plant under Bang-Bang control, the output slope $\dot{y}(t)$ changes sign from negative to positive exactly at $t = t_{\mathrm{sw}} + L$ after a rising switch ($U_{\min} \to U_{\max}$), because only then does the new input value reach the plant through the dead-time element. The dead time is therefore estimated as

$$\hat{L} = t_{\mathrm{sign-change}} - t_{\mathrm{sw}}, \tag{15}$$

averaged over the last $N_{\min}$ rising-switch events.

Time constant estimate. At $t = t_{\mathrm{sw,fall}} + \hat{L}$ after a falling switch ($U_{\max} \to U_{\min}$), the FOPDT step-response slope for $U_{\min} = 0$ is

$$\dot{y}\big|_{t_{\mathrm{sw,fall}}+\hat{L}} = \frac{K \cdot U_{\min} - y_L}{T} = \frac{-y_L}{T}, \tag{16}$$

where $y_L = y(t_{\text{sw,fall}} + \hat{L})$. Solving for T:

$$\hat{T} = \frac{y_L}{|\dot{y}|_{t_{\text{sw,fall}}+\hat{L}}}, \tag{17}$$

averaged over the last $N_{\min}$ falling-switch events.

Gain estimate. At $t = t_{\text{sw,rise}} + \hat{L}$ after a rising switch, the slope is

$$\dot{y}\big|_{t_{\text{sw,rise}}+\hat{L}} = \frac{K \cdot U_{\max} - y_L}{T}, \tag{18}$$

hence

$$\hat{K} = \frac{\hat{T} \cdot \dot{y}_{L,\uparrow} + y_{L,\uparrow}}{U_{\max}}, \tag{19}$$

averaged over the last $N_{\min}$ rising-switch events.

Remark 3.1. (17) *is exact for* $U_{\min} = 0$. *For* $U_{\min} \neq 0$, *an iterative refinement involving* $K \cdot U_{\min}$ *is required; this case is not treated here.*

3.4 Phase 4: PID Synthesis

With the identified FOPDT model $(\hat{K}, \hat{T}, \hat{L})$, apply the IMC-PID formulas (10)–(12) with $\lambda = \hat{L}$:

$$\boxed{K_{\text{p}} = \frac{\hat{T}}{\hat{K}(2\hat{L})}, \quad T_{\text{i}} = \hat{T}, \quad T_{\text{d}} = \frac{\hat{L}}{2}.} \tag{20}$$

Switch the controller from Bang-Bang mode to PID mode. No additional plant experiment is required.

3.5 Implementation Notes

- The procedure is implemented in the Python module `analysis/parameter_estimation.py` see the repository for source code.
- All simulation and figure generation scripts are available in the `python/` directory; see `README.md` for usage.
- The same procedure can be implemented on a PLC or embedded controller by storing a rolling buffer of the last 3–5 oscillation cycles and computing the statistics online.

4 Simulation Results

4.1 Test Plant and Setup

Validation uses a benchmark FOPDT plant

$$G(s) = \frac{0.5\,e^{-20s}}{120s+1} \quad \left[K_0{=}0.5\,^\circ\mathrm{C}\,\%^{-1},\ T_0{=}120\,\mathrm{s},\ L_0{=}20\,\mathrm{s}\right], \tag{21}$$

with $U_{\max} = 100\,\%$, $U_{\min} = 0\,\%$, hysteresis $d = 0.3\,^\circ\mathrm{C}$, setpoint $r = 20\,^\circ\mathrm{C}$, and simulation step $\Delta t = 0.5\,\mathrm{s}$.

4.2 Bang-Bang Limit Cycle

Figure 6 shows the stationary limit cycle. Averaged over five complete cycles: $T_\mathrm{u} = 82.6\,\mathrm{s}$, $A_\mathrm{u} = 8.29\,^\circ\mathrm{C}$.

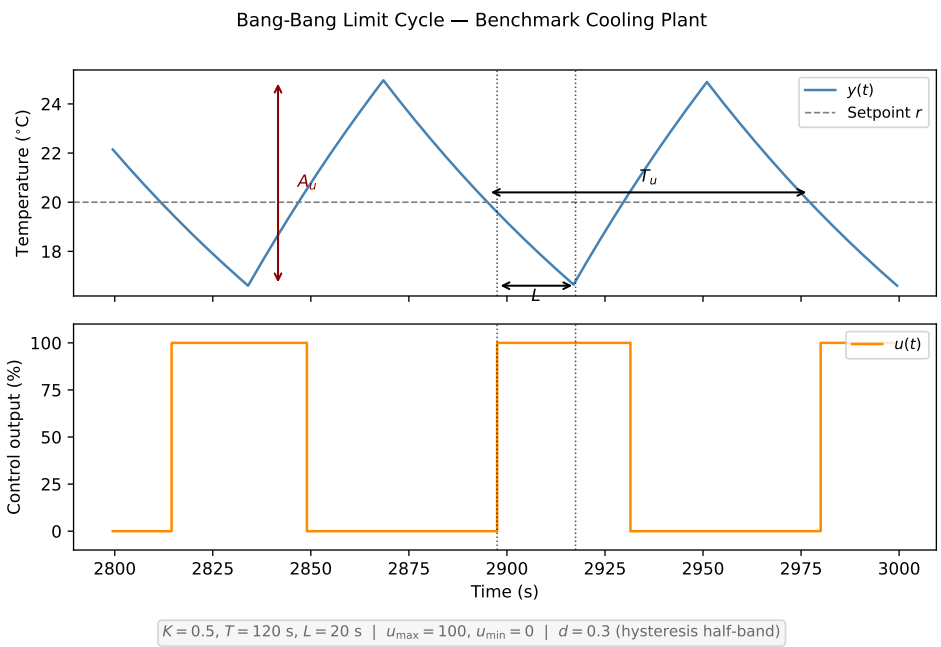

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Figure 6: Stationary limit-cycle oscillation under Bang-Bang control. *Top*: plant output $y(t)$ and setpoint. *Bottom*: manipulated variable $u(t)$.

4.3 Identified FOPDT Parameters

Table 1 compares the KZV estimates to the true plant values. The gain and time constant are recovered exactly; the dead-time error of 2.5 % is due to

the discrete time step.

Table 1: KZV-identified vs. true FOPDT parameters.

Parameter	True	KZV estimate	Error (%)
K [°C %$^{-1}$]	0.500	0.500	0.0
T [s]	120.0	120.0	0.0
L [s]	20.0	19.5	2.5

4.4 Closed-Loop Performance

Figure 7 compares step responses of three PID configurations: **KZV-IMC** (parameters from (20)), **ZN** (classical Ziegler–Nichols on the identified model), and **Ideal IMC** (tuned on the exact plant — upper bound).

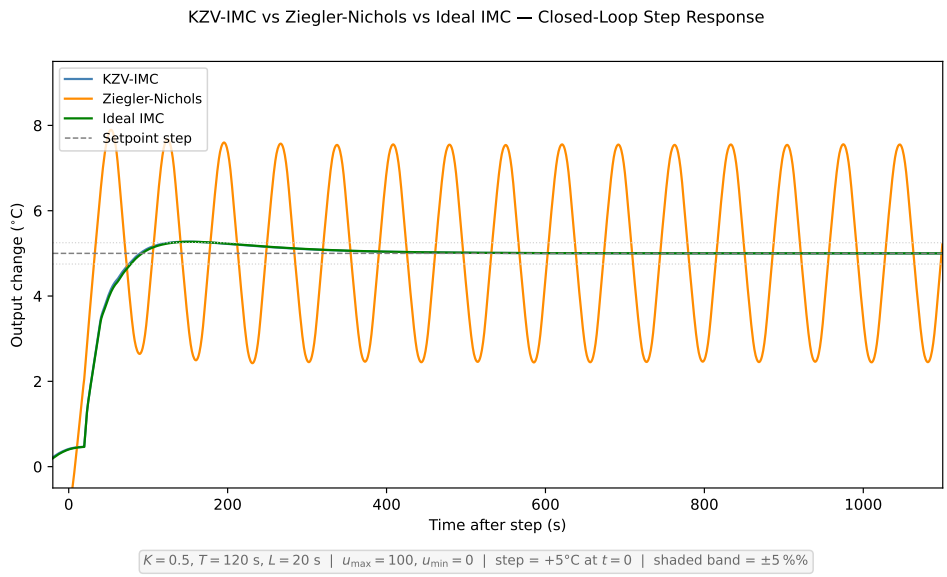

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Figure 7: Closed-loop step responses for a $\Delta r = 5\,°\mathrm{C}$ step. Dashed band: ±5% settling criterion.

KZV-IMC achieves 5.5 % overshoot and an IAE within 2 % of the ideal IMC result, while ZN gives 57.9 % overshoot and an eightfold larger IAE and does not settle within the 1200 s simulation horizon.

Table 2: Performance metrics for the $\Delta r = 5\,°\mathrm{C}$ step.

Metric	KZV-IMC	ZN	Ideal IMC
Overshoot (%)	5.5	57.9	5.4
Settling time (s)	76	—	79
IAE (°C s)	228	1834	232

4.5 Robustness Across Operating Points and Plant Ratios

Figure 8 demonstrates the KZV's robustness: PID parameters are re-identified from a Bang-Bang experiment at each combination of setpoint ($r \in \{15, 20, 25\}$ °C) and L/T ratio (0.1, 1.0, 10.0), and the resulting step responses are shown. Across all nine scenarios the KZV-IMC controller delivers consistent, low-overshoot tracking.

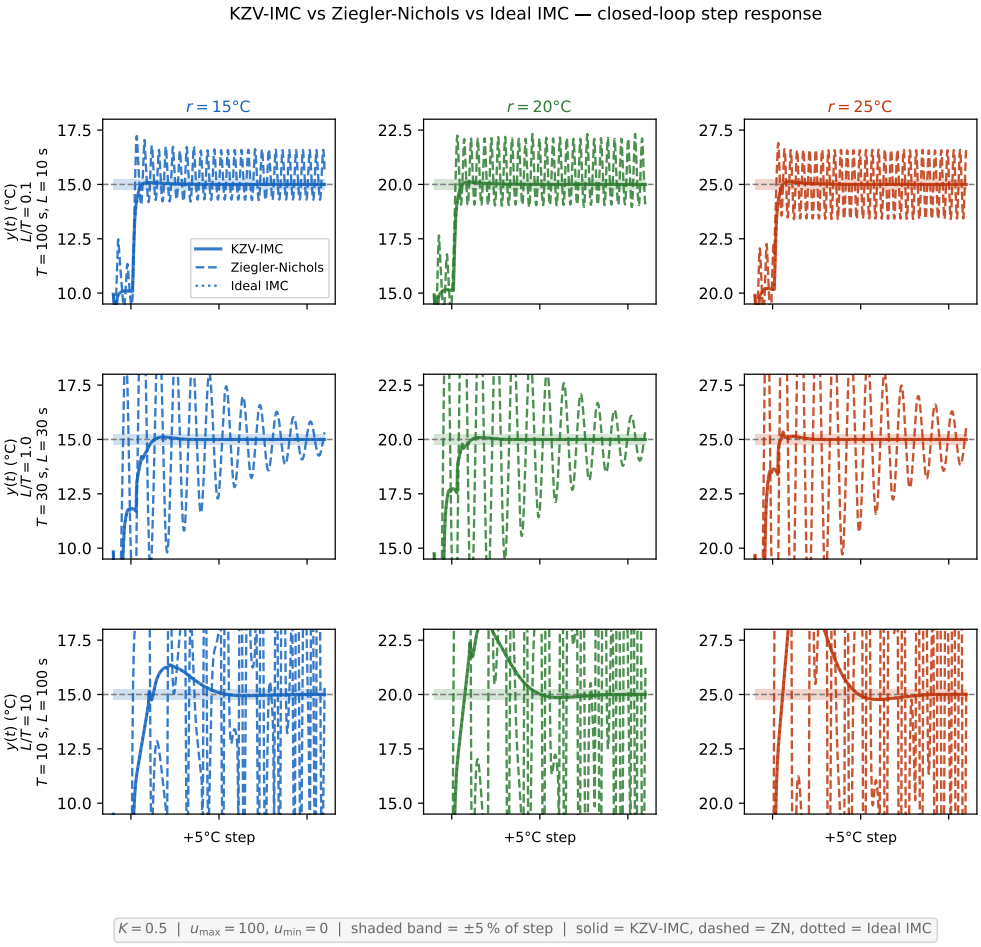

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Figure 8: KZV-IMC step responses for three setpoints ($r \in \{15, 20, 25\}$ °C, rows) and three L/T ratios (columns). The method delivers consistent closed-loop behaviour across all operating conditions.

5 Discussion

The describing function approximation used in section 2.2 assumes that the plant sufficiently attenuates higher harmonics of the relay switching signal, so that the fundamental frequency component dominates. This assumption holds well for plants with a roll-off of at least $-40\,\mathrm{dB/decade}$ above ω_u. For thermal processes (which typically roll off much more steeply), the approximation is highly accurate.

For plants with resonances near ω_u, the describing function prediction can be less accurate. In such cases, it is recommended to add a low-pass filter in series with the measurement before computing A_u and T_u.

5.1 Robustness to Measurement Noise

Measurement noise directly affects the accuracy of A_u and, more critically, the switching instants used to compute T_u. A practical mitigation strategy is to:

1. Use an adequate hysteresis $d \gg \sigma_n$, where σ_n is the standard deviation of the measurement noise.
2. Average over at least five oscillation cycles (as prescribed in section 3.2).
3. Apply a moving-average pre-filter with window length $w = T_\mathrm{u}/10$ before detecting zero-crossings.

5.2 Extension to SOPDT Plants

The FOPDT identification described in this paper is intentionally conservative: it maps any higher-order plant onto a simple FOPDT model. For plants where a second time constant is significant, the estimated dead time $\hat{L}$ absorbs part of the lag contributed by T_2, leading to a slightly conservative (i.e. detuned) PID. This is generally desirable for robust industrial control. An extension using a second-order relay feedback test [8] is left for future work.

5.3 Limitations

- The procedure requires the plant to be stable and to sustain a limit cycle under Bang-Bang control. Integrating or unstable plants require a modified relay feedback scheme.

- Nonlinear plant behaviour (e.g. hysteresis, saturation) may cause the limit cycle to be asymmetric, leading to biased estimates.
- The method is derived for SISO systems. Multi-variable extensions are not considered here.

5.4 Comparison with Related Methods

The KZV is most closely related to the relay feedback tuning method of Åström and Hägglund [1], which also uses a relay (Bang-Bang without hysteresis) to excite a sustained oscillation. The two methods share the same underlying experiment but make different trade-offs:

- **Åström & Hägglund** read a single point on the Nyquist curve (K_{u}, ω_u) from the limit cycle via describing-function analysis. Because this requires essentially no assumption about the plant structure, the method is broadly applicable across very different process types. PID parameters are then obtained via Ziegler–Nichols-style rules, which do not explicitly account for dead time.
- **KZV** extracts three time-domain slope measurements from the same limit cycle to estimate the full FOPDT triple $(\hat{K}, \hat{T}, \hat{L})$ and tunes via IMC, which exposes a robustness knob λ and treats dead time as a first-class parameter. The price for this richer model is two-fold: (i) the plant must be reasonably well approximated by FOPDT, and (ii) slope estimates are more sensitive to measurement noise than amplitude/period estimates.

The two approaches are therefore best viewed as *complementary*. For general-purpose auto-tuning across heterogeneous plants, classical relay feedback remains the more conservative default. For the FOPDT-friendly process class targeted in this work — in particular thermal and cooling systems with non-negligible dead time — the KZV's explicit dead-time estimate and IMC tuning typically yield a more transparent and easier-to-tune result from the same experimental data.

6 Conclusion

The method exploits the stationary limit-cycle oscillation that a Bang-Bang controller naturally produces on a stable plant. Using describing function analysis, three FOPDT model parameters (gain, time constant, dead time)

are identified from the measured oscillation period T_u, amplitude A_u, and the known relay switching levels. These parameters are then fed into IMC-based PID tuning formulas to obtain K_p, T_i, and T_d.

Simulation results on the benchmark FOPDT plant (section 4) demonstrate that KZV-IMC achieves 5.5 % overshoot and an IAE within 2 % of the theoretically optimal IMC controller, while Ziegler–Nichols tuning yields 57.9 % overshoot and more than eight times the IAE.

The main advantages of the KZV over classical methods are:

- **No additional experiment needed**: the Bang-Bang phase already present during commissioning is sufficient.
- **Dead-time estimation**: unlike basic ZN relay feedback, the KZV provides a separate estimate of dead time, enabling more robust IMC-style tuning.
- **Simple implementation**: all computations involve closed-form expressions and can be executed on standard PLC hardware.

Future Work

Open directions include:

- Experimental validation on a physical cooling test bench.
- Extension to second-order identification using multi-frequency relay tests.
- Online (recursive) adaptation of the PID parameters as the plant ages or operating conditions change.
- Integration with auto-tuning firmware for industrial controllers.

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